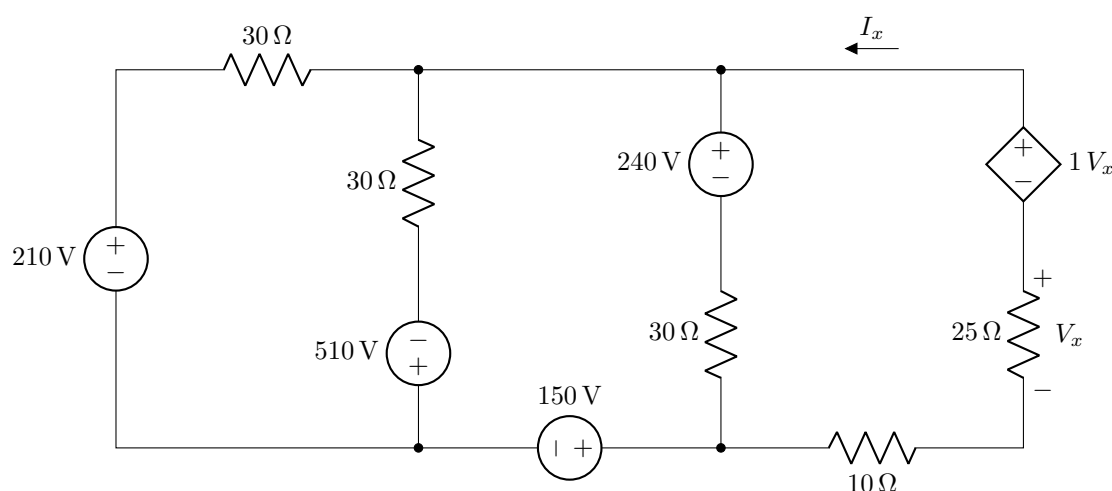
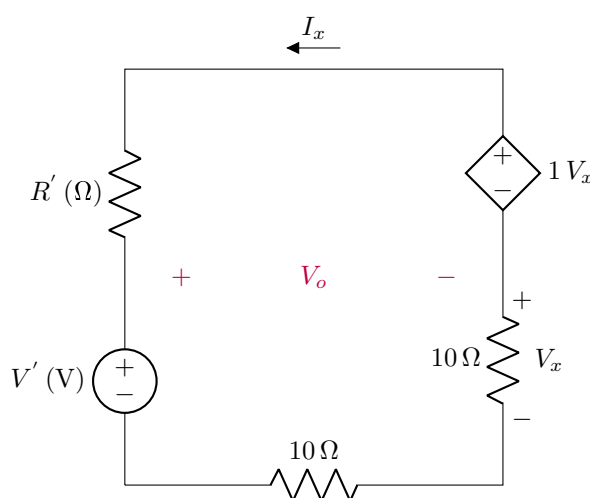


- ✓ No washroom breaks. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, the **answer script** must be submitted to the form.
- ✓ All **4 questions** are compulsory. Marks allotted for each question are mentioned beside each question.
- ✓ Using any external smart device during the exam is strictly prohibited.
- ✓ Symbols have their usual meanings.

■ **Question 1 of 4 [CO2] [15 marks]**



Circuit 1

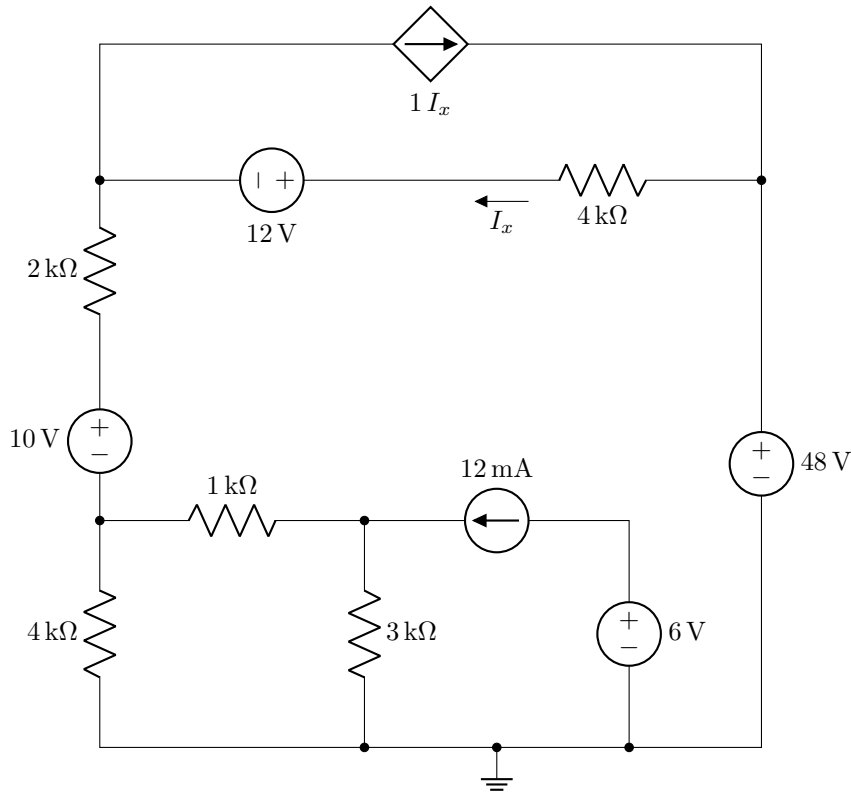


Circuit 2

Apply Source Transformation to answer the following queries–

- (a) [8 marks] Reduce the Circuit 1 so that it takes the form of Circuit 2. Determine the values of V' and R' .
- (b) [4 marks] Determine the values of V_x and I_x .
- (c) [3 marks] Determine the value of V_o from Circuit 2.

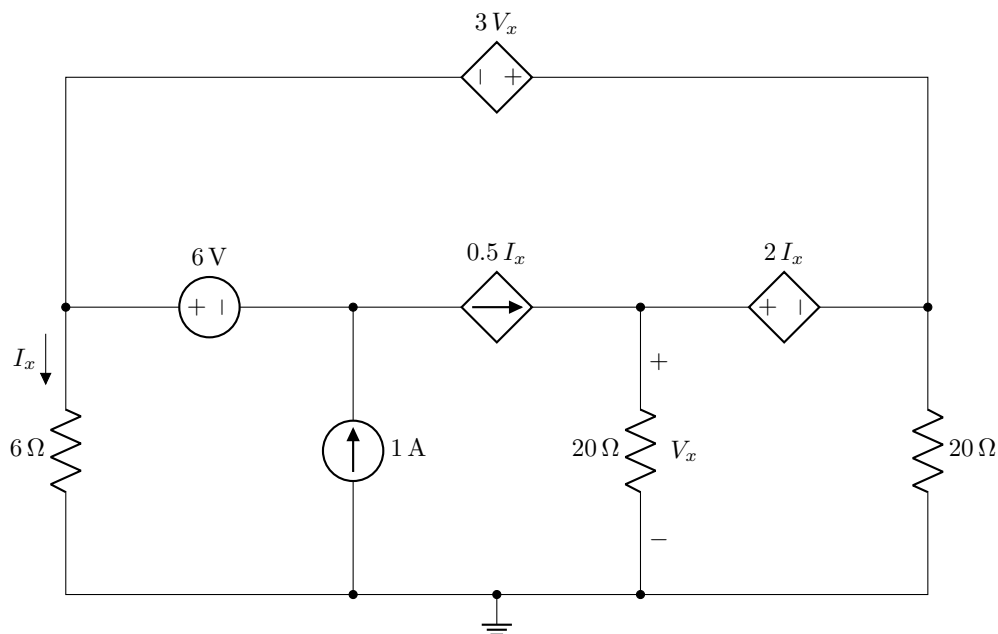
■ Question 2 of 4 [CO3] [15 marks]



Apply Nodal/Mesh analysis to answer the following questions–

- [12 marks] Find all the node voltages/mesh currents in the circuit shown above. Note that, depending on the analysis method you are applying, you have to determine either the mesh currents or the node voltages, not both.
- [3 marks] Determine the power of the $1 I_x$ dependent current source (with appropriate $\pm$ sign). Also mention, whether the source is supplying or consuming the power.

■ Question 3 of 4 [CO3] [15 marks]

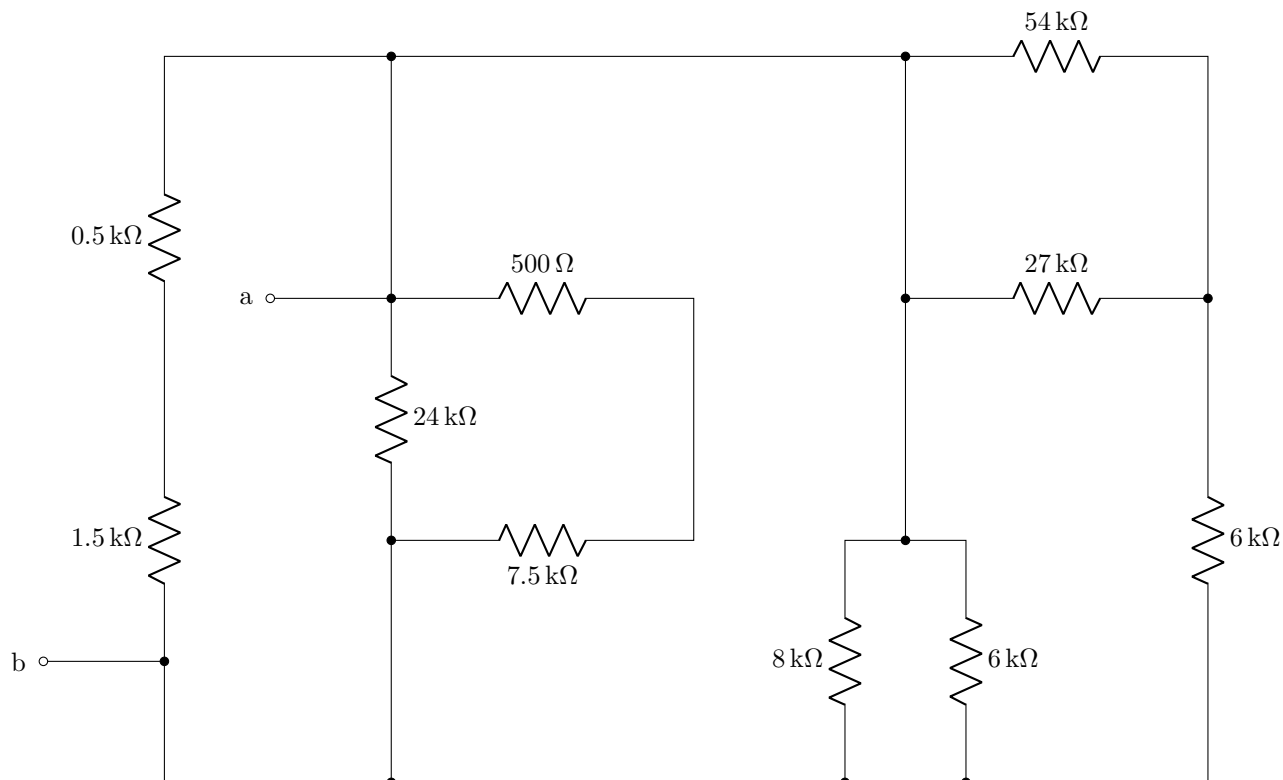


Apply Nodal/Mesh analysis to answer the following questions–

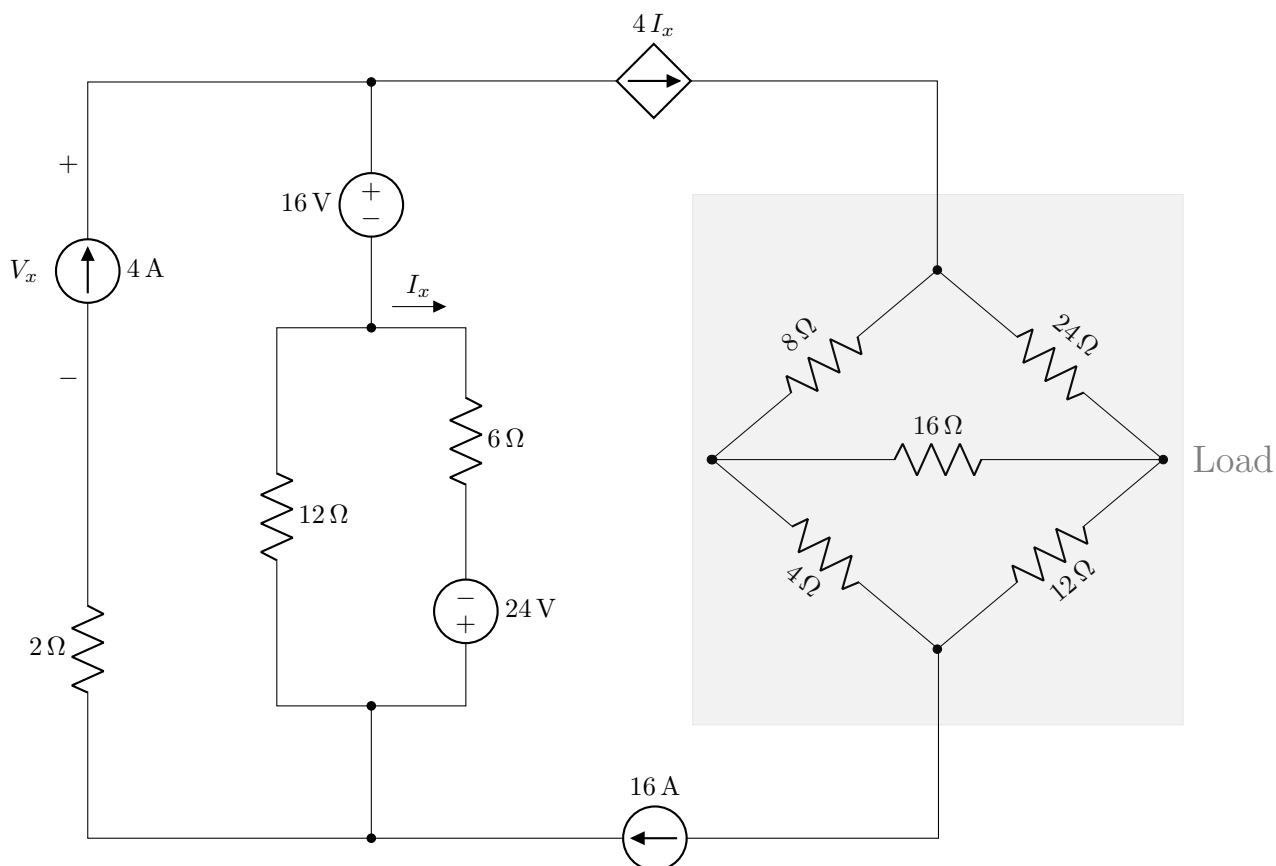
- [12 marks] Find all the node voltages/mesh currents in the circuit shown above. Note that, depending on the analysis method you are applying, you have to determine either the mesh currents or the node voltages, not both.
- [3 marks] Determine V_x and I_x .

■ Question 4 of 4 [CO3] [15 marks]

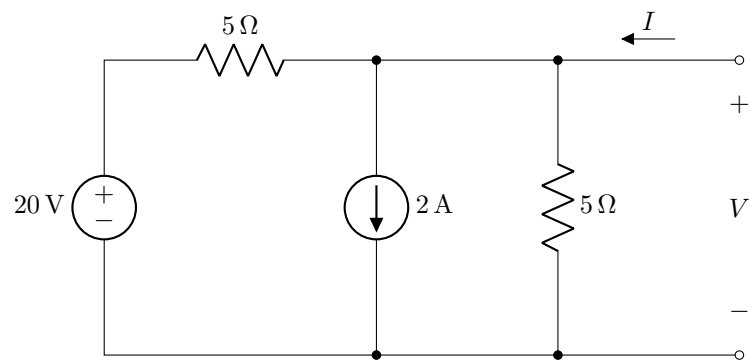
- (a) [6 marks] Determine the equivalent resistance between terminals a and b for the circuit shown below.



- (b) [3 marks] For the circuit shown below, if $I_x = 4\text{ A}$, determine the value of V_x .



- (c) (i) [4 marks] Approximately draw the $I - V$ characteristics of the following circuit. Label the axes and the intersecting points clearly.



- (ii) [2 marks] Now draw the $I - V$ characteristics of the same circuit in the same grid if the 20 V voltage source and the 2 A current source are turned off.